

PRACTICAL NO. 12

FEASIBILITY OF A TECHNO-COMMERCIAL BUSINESS

Aim: Preparation of Case Study Report on “Feasibility of a Techno-Commercial Business” using SolarSmart Irrigation – Solar-Powered Smart Irrigation System as a case study.

1. Introduction

A techno-commercial feasibility study is carried out to determine whether a proposed business is technically practical and commercially viable. It considers the technology, market demand, resources, investment, operating costs, expected revenue, risks and support required before starting the enterprise.

This case study evaluates the feasibility of SolarSmart Irrigation, a proposed solar-powered smart irrigation system designed to help farmers automate irrigation while reducing dependence on conventional electricity. The study combines technical feasibility with basic commercial assessment.

2. Selected Business

Selected Business: SolarSmart Irrigation – Solar-Powered Smart Irrigation System

Business Concept: The proposed enterprise will develop and supply a solar-powered irrigation solution consisting of a solar panel, controller, water pump interface, soil-moisture sensing and basic automated control. The system is intended for farms where reliable irrigation and electricity access are important concerns.

3. Major Factors Affecting Techno-Commercial Feasibility

Factor	Assessment Area	Importance
Technology	Solar power, sensors, controller, pump interface and reliability	Determines whether the product can perform the intended function.
Market	Farmers, irrigation need, competing solutions and customer demand	Determines potential sales and market acceptance.
Investment	Product development, components, tools, testing and working capital	Determines initial financial requirement.
Operating Cost	Components, assembly, labour, service and maintenance	Affects selling price and profitability.
Selling Price	Affordable price compared with customer benefits and alternatives	Affects demand and revenue.
Infrastructure	Workspace, testing facilities, suppliers and service network	Supports production and after-sales service.
Risk	Technical failure, component price changes, seasonal demand and competition	Helps identify measures required for business sustainability.

4. SolarSmart Irrigation – About the Business

SolarSmart Irrigation is considered as a small technology-oriented enterprise. Its proposed product combines renewable-energy generation with basic automation for agricultural irrigation.

The technical system may include a solar photovoltaic panel, charge/controller unit, water pump, soil-moisture sensor, microcontroller-based control unit, wiring and protective enclosure. Depending on customer requirements, the system can be supplied as a basic automatic irrigation kit or as a more advanced connected solution.

The commercial objective is to offer a practical irrigation solution with an affordable initial price, reduced dependence on grid electricity and a serviceable design.

5. Technical Feasibility

The technical feasibility of the proposed business depends on whether the product can be designed, assembled, tested and maintained using reasonably available components and skills.

Technical Requirement	Feasibility Consideration
Solar panel	Selected according to pump load, sunlight availability and required operating hours.
Pump and irrigation connection	Standard pump and irrigation components can be integrated with suitable control arrangements.
Soil-moisture sensing	Sensor input can be used to start or stop irrigation according to preset moisture levels.
Controller	A microcontroller or suitable control unit can automate the irrigation sequence.
Power management	Battery/charge-control arrangements may be included where operation outside sunlight hours is required.
Testing and maintenance	Prototype testing, electrical protection and periodic sensor/pump checks are required.

6. Commercial Feasibility

Commercial feasibility considers whether sufficient customers are likely to purchase the product at a price that can cover operating expenses and provide a reasonable return.

Commercial Aspect	Case Study Assessment	Result
Customer Need	Irrigation is an essential farm activity and automation can reduce manual intervention.	Favourable
Target Market	Small and medium farms requiring efficient irrigation and suitable solar options.	Potential market
Competition	Conventional electric/diesel pumps and other solar irrigation systems are alternatives.	Requires differentiation
Pricing	Price should balance affordability with component and service costs.	Feasible with cost control
Revenue	Revenue can come from system sales, installation and maintenance/service.	Multiple revenue sources
Scalability	Standardized modules can support expansion into more farms and locations.	Favourable

7. Illustrative Financial Feasibility

The following figures are illustrative academic assumptions for evaluating feasibility. They are not the actual internal cost structure of any existing enterprise.

Particular	Illustrative Amount (₹)	Basis
Initial product development and setup	3,00,000	Prototype, tools, testing and basic setup
Estimated cost per system	18,000	Components, assembly and testing
Illustrative selling price per system	25,000	Target commercial price
Contribution per system	7,000	Selling price – variable cost
Annual fixed operating cost	7,00,000	Staff, workspace, administration and other fixed expenses
Expected annual sales	200 systems	Illustrative first-year target

8. Basic Commercial Analysis

At an illustrative selling price of ₹25,000 and variable cost of ₹18,000 per system, the contribution per system is ₹7,000.

Estimated annual contribution at 200 systems = $200 \times ₹7,000 = ₹14,00,000$.

After deducting illustrative annual fixed operating cost of ₹7,00,000, the estimated operating surplus is ₹7,00,000 before considering taxes, financing cost and depreciation.

Indicative break-even quantity = $₹7,00,000 \div ₹7,000 = 100$ systems per year.

Thus, under these assumptions, the illustrative sales target of 200 systems is above the break-even level. The result indicates commercial potential, but actual feasibility would require detailed market validation, supplier quotations, field trials and a complete project report.

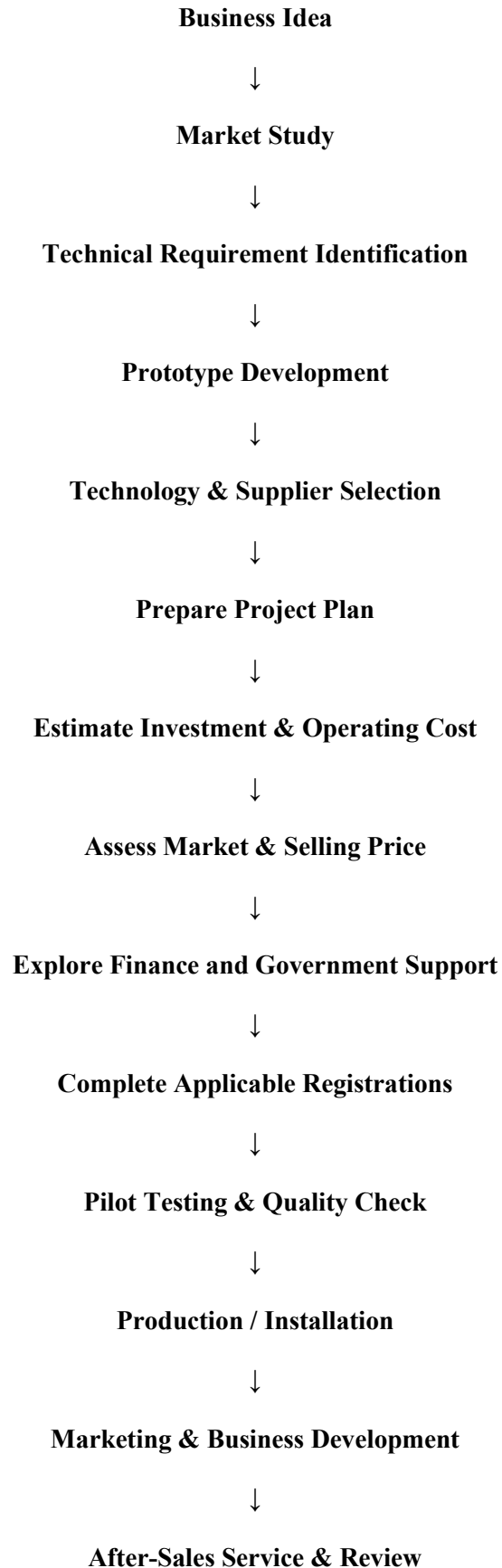
9. Information Required Before Starting the Business

Requirement	Information to be Collected
Business Idea	Product configuration, customer problem and proposed value.
Market	Target farmers, demand, competition and customer willingness to pay.
Technology	Solar capacity, pump specifications, sensors, controller and reliability.
Location	Suitable place for assembly, testing, storage and service.
Raw Materials	Sources, prices and availability of panels, pumps, sensors and electronics.
Finance	Initial investment, working capital, funding options and cash-flow requirement.
Registration	Udyam and other registrations/approvals applicable to the business.
Training	Technical, entrepreneurship and installation/service skills.
Government Schemes	Applicable renewable-energy, MSME or agriculture-related support.
Marketing	Dealer network, demonstrations, promotion and sales channels.
Legal Requirements	Applicable electrical, safety, tax and business requirements.

10. Support Required by SolarSmart Irrigation

Requirement	Possible Support Source	Purpose
Entrepreneurship training	MCED / MSME	Business planning and entrepreneurship skills
Technical development	Technical institutions / training centres	Prototype development and testing
MSME registration	Udyam Registration	Formal MSME registration where applicable
Finance-related support	Banks / applicable schemes	Capital and working-capital support
Market support	Applicable MSME/agriculture programmes	Market access and promotion
Local facilitation	DIC	Local guidance and enterprise facilitation
Project information	MCED / MSME / relevant departments	Information for preparing the project plan

11. Suggested Steps for Establishing the Techno-Commercial Business



12. Key Findings

1. A techno-commercial feasibility study combines technical practicality with market and financial considerations.
2. SolarSmart Irrigation has a technically achievable concept because its major components can be integrated into a controlled irrigation system.
3. The proposed business has potential demand where farmers seek reliable irrigation and reduced dependence on conventional electricity.
4. The commercial result depends strongly on component costs, selling price, annual sales volume and service expenses.
5. The illustrative financial analysis shows a break-even level of about 100 systems per year under the stated assumptions.
6. Market testing, prototype trials, supplier quotations and customer feedback are necessary before commercial launch.
7. The entrepreneur should check the eligibility and current guidelines of each applicable scheme before applying.

13. Conclusion

The feasibility of a techno-commercial business depends on the successful combination of technology, market demand, cost management, finance and implementation capability. The SolarSmart Irrigation case study indicates that a solar-powered smart irrigation system can be considered technically feasible and commercially promising when designed for a clearly defined customer segment.

However, the financial figures used in this case study are illustrative academic assumptions and should not be treated as actual market quotations or guaranteed returns. Before establishing the enterprise, the entrepreneur should conduct detailed market research, obtain supplier quotations, validate the prototype, estimate actual operating costs and examine applicable registrations and support schemes.

Therefore, a systematic techno-commercial feasibility study is an important step before committing significant resources to a new technology-based business.

15. References

1. MSBTE, Entrepreneurship Development and Startups, Course Code 315002.
2. Maharashtra Centre for Entrepreneurship Development (MCED).
3. Ministry of Micro, Small and Medium Enterprises, Government of India.
4. Udyam Registration Portal, Government of India.
5. Relevant renewable-energy and agriculture-related government information sources.